

Beginner Lab Guide: Create an EC2 Instance from the AWS Console and Install Terraform on Ubuntu

1. Create an Ubuntu EC2 Instance from the AWS Console

Step 1: Sign in and open the EC2 console

1. Open the AWS Management Console.
2. In the search bar, type **EC2**.
3. Select **EC2** from the service results.
4. Confirm the AWS Region in the top-right corner. For training, ask all learners to use the same Region, such as **Asia Pacific (Mumbai)**, unless your class uses a different Region.

Screenshot 1: AWS Console search bar showing the EC2 service selected.

Step 2: Start the launch instance wizard

1. On the EC2 dashboard, choose **Instances**.
2. Choose **Launch instances**.
3. In the **Name and tags** section, enter a clear name such as **terraform-beginner-lab**.

Screenshot 2: Launch instance page showing the instance name field.

Step 3: Select Ubuntu as the operating system

1. Under **Application and OS Images**, choose **Quick Start**.
2. Select **Ubuntu**.
3. Choose a recent Ubuntu Server LTS AMI, such as Ubuntu Server 24.04 LTS or the latest available LTS option in your Region.
4. Keep the default architecture unless your instructor asks otherwise.

Beginner note: An AMI is a template used to create a virtual server. Choosing Ubuntu means the EC2 instance will start with the Ubuntu operating system already installed.

Screenshot 3: Application and OS Images section with Ubuntu selected.

Step 4: Choose instance type

1. In **Instance type**, choose **t2.micro** or **t3.micro**, depending on availability and free tier eligibility in your account.
2. For this beginner lab, do not choose large instance types.

Screenshot 4: Instance type dropdown showing a small instance type selected.

Step 5: Create or select a key pair

1. In the **Key pair** section, choose an existing key pair or choose **Create new key pair**.

2. Enter a key pair name such as **terraform-lab-key**.
3. Choose **RSA** and **.pem**.
4. Choose **Create key pair**.
5. The private key file downloads automatically. Save it in a safe folder and do not share it.

Beginner note: The key pair works like a secure password file. If you lose the private key file, you may not be able to connect to the instance using SSH.

Screenshot 5: Create key pair dialog showing key name, RSA, and .pem selected.

Step 6: Configure network and security group

1. In **Network settings**, allow AWS to create or select a VPC and subnet for the lab, unless your instructor provides a specific network.
2. Enable **Auto-assign public IP** if shown.
3. For **Firewall**, select **Create security group**.
4. Add a rule for **SSH**, protocol **TCP**, port **22**.
5. For source, choose **My IP** instead of Anywhere.

Security warning: Do not use **Anywhere 0.0.0.0/0** for SSH in beginner labs unless your instructor specifically explains the risk. Restricting SSH to **My IP** is safer.

Screenshot 6: Security group rule showing SSH on port 22 with source set to My IP.

Step 7: Configure storage and launch

1. In **Configure storage**, keep the default root volume size unless your instructor asks for a different value. For this lab, 8 GB or more is usually enough.
2. Review the summary panel.
3. Choose **Launch instance**.
4. Wait until the instance state becomes **Running**.
5. Wait until status checks show that the instance is ready.

Screenshot 7: Instance summary page showing the instance state as Running.

4. Connect to the Ubuntu EC2 Instance

Option A: Connect using EC2 Instance Connect in the browser

1. Select the running instance.
2. Choose **Connect**.
3. Select the **EC2 Instance Connect** tab.
4. Confirm the username is **ubuntu**.

5. Choose **Connect**.

Screenshot 8: EC2 Connect page showing username ubuntu and the Connect button.

Option B: Connect using SSH from your computer

From the EC2 console, copy the instance **Public IPv4 address** or **Public IPv4 DNS**. Then open a terminal on your computer and go to the folder where your downloaded key file is saved.

Linux/macOS/Windows PowerShell command examples:

```
chmod 400 terraform-lab-key.pem
ssh -i terraform-lab-key.pem ubuntu@<PUBLIC-IP-OR-DNS>
```

Beginner note: Replace **<PUBLIC-IP-OR-DNS>** with the actual public IPv4 address or public DNS name from your EC2 instance details.

5. Install Terraform on Ubuntu

Run the following commands inside the connected Ubuntu EC2 terminal.

Step 1: Update Ubuntu package information

```
sudo apt-get update
```

Purpose: This refreshes the list of software packages available to Ubuntu.

Step 2: Install required tools

```
sudo apt-get install -y gnupg software-properties-common wget
```

Purpose: These tools allow Ubuntu to verify and use HashiCorp's official software repository.

Step 3: Add the HashiCorp GPG key

```
wget -O- https://apt.releases.hashicorp.com/gpg | gpg --dearmor | sudo
tee /usr/share/keyrings/hashicorp-archive-keyring.gpg > /dev/null
```

Purpose: The GPG key helps Ubuntu confirm that Terraform packages come from HashiCorp and have not been tampered with.

Step 4: Add the HashiCorp APT repository

```
echo "deb [arch=$(dpkg --print-architecture) signed-
by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(grep -oP
```

```
'(?<=UBUNTU_CODENAME=).*' /etc/os-release || lsb_release -cs) main" |  
sudo tee /etc/apt/sources.list.d/hashicorp.list
```

Purpose: This tells Ubuntu where to download Terraform packages from.

Step 5: Update package information again

```
sudo apt-get update
```

Step 6: Install Terraform

```
sudo apt-get install -y terraform
```

Step 7: Verify Terraform installation

```
terraform -version  
terraform -help
```

Expected result: The terminal displays the installed Terraform version and a list of Terraform commands.